ARDA FAZLA

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EDUCATION

Purdue University — West Lafayette, IN
Ph.D. Electrical and Computer Engineering — Advisor: Prof. Abolfazl Hashemi — 2024 – 2029 (Expected)

Bilkent University — Ankara, Turkey
M.S. Electrical and Electronics Engineering — Advisor: Prof. Süleyman Serdar Kozat — 2021 – 2024

Middle East Technical University — Ankara, Turkey
B.S. Electrical and Electronics Engineering — 2017 – 2021

RESEARCH INTERESTS

- **Optimization Theory:** Nonconvex stochastic optimization, lower bounds, variance reduction methods
- **LLM Pretraining:** Efficient large-scale training, adaptive and spectral optimizers
- **RL & Post-Training:** RL-based LLM post-training, LLM distillation

EXPERIENCE

Research Assistant, Purdue University — Aug 2024 – Present

- **LLM pretraining at scale** (under review, ICLR 2027): Measured gradient-noise growth during LLM pretraining and used the results to introduce the generalized Blum–Gladyshev (BG-a) noise model. Based on this analysis, designed an **adaptive batch-size scheduler** for LLM pretraining that achieves lower validation loss than small- and large-batch training at matched token budgets. Proved an information-theoretic lower bound of $\Omega(\epsilon^{-(4+a)})$ for convergence under the proposed BG-a noise model with matching upper bounds.
- **Stochastic optimization** (NeurIPS 2026; under review, ICLR 2027): Developed PASTA and SAUCE for nonconvex learning under unbounded variance. Established oracle-complexity lower bounds and **minimax-optimal** rates for SGD, and provided convergence guarantees for normalized momentum and variance-reduced (STORM) methods.
- **LLM fine-tuning** (under review, AISTATS 2027): Contributed to constructing robust adaptive Langevin methods with non-asymptotic guarantees under state-dependent heavy-tailed noise. In **Bayesian fine-tuning of LLMs and vision-language models**, our proposed method outperforms state-of-the-art baselines.
- **Markovian stochastic approximation** (under review, ICLR 2027): Contributed to constructing a variance-reduced Markovian Halpern method for fixed-point problems with samples from a continuing Markovian trajectory, with $\tilde{O}(\epsilon^{-3})$ sample complexity in general Banach spaces.
- **Data-efficient deep learning** (UAI 2026): Proposed a two-stage coreset selection algorithm for datasets with spurious correlations that separates the learning dynamics of core and spurious features. Achieved up to **27% improvement** over standard ERM using only **10% of the training data**, and outperformed state-of-the-art invariant learning methods.

Data Scientist, Databoss Security & Analytics Inc. — Aug 2021 – May 2024

- Built a large-scale forecasting framework for hourly energy output of wind turbines across Turkey and Europe, combining deep learning (ConvLSTM), gradient boosting (LightGBM), statistical (SARIMAX), and ensemble models. The deployed product improved short-term accuracy by **21%** and long-term accuracy by **8%**.
- Built a daily natural gas demand forecasting framework for nonstationary multivariate time series with automatic extraction of key indicators. Deployed at multiple natural gas production companies.

Machine Learning Engineer, METU Center for Image Analysis (OGAM) — Jun 2020 – Jul 2021

- Developed a real-time small-object tracking application for UAV data that uses temporal information from detector outputs, with **6% better** performance than YOLOv3.
- Built a multi-hypothesis tracking algorithm and a semi-automatic annotation tool that reduced mislabeled data by an average of **13%** (ICCV Workshops 2021).

SELECTED PUBLICATIONS

Published

- A. Upadhyay, **A. Fazla**, A. Hashemi. "Beyond Bounded Variance: Variance-Reduced Normalized Methods for Nonconvex Optimization under Blum-Gladyshev Noise." ***NeurIPS 2026***. [PDF]
- **A. Fazla**, A. Hashemi. "Mitigating Spurious Correlations with Memorization-Guided Dataset De-Biasing." ***UAI 2026***. [PDF]
- S. F. Tekin, **A. Fazla**, S. S. Kozat. "Numerical Weather Forecasting using Convolutional-LSTM with Attention and Context Matcher Mechanisms." ***IEEE Trans. Geoscience and Remote Sensing, 2024***. [PDF] [Code]
- **A. Fazla**, M. E. Aydin, S. S. Kozat. "Time-Aware and Context-Sensitive Ensemble Learning for Sequential Data." ***IEEE Trans. Artificial Intelligence, 2023***. [PDF] [Code]

Under Review

- **A. Fazla**, A. Upadhyay, E. Kaya, M. Sahin, A. Hashemi. "Why Does Adaptive Batching Help LLM Pretraining? A Perspective from Unbounded Variance." ***ICLR 2027 (under review)***. [PDF]
- **A. Fazla**, E. Kaya, A. Upadhyay, A. Hashemi. "Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance." ***ICLR 2027 (under review)***. [PDF]
- M. Sahin, E. Kaya, **A. Fazla**, B. Sharif, A. Hashemi. "Robust Langevin Sampling under State-Dependent Heavy-Tailed Noise with Applications to LLM Fine-Tuning." ***AISTATS 2027 (under review)***. [PDF]
- E. Kaya, **A. Fazla**, M. Sahin, A. Hashemi. "A Banach-Space Theory of Markovian Halpern Iteration for Non-Expansive Maps." ***ICLR 2027 (under review)***. [PDF]

TECHNICAL SKILLS

- **Programming:** Python, C/C++, MATLAB, R
- **ML Frameworks:** PyTorch, JAX, TorchTitan, Hugging Face
- **LLM Training:** Multi-GPU and multi-node training (DDP/FSDP)
- **Post-Training:** SFT, RLHF and RLVR (DPO, PPO, GRPO, GSPO), LLM distillation (OPD)
- **Infrastructure:** SLURM/HPC clusters, Docker, Git, Weights & Biases

HONORS & AWARDS

- **Elmore PhD Fellowship**, Elmore Family School of ECE, Purdue University 2024 – 2029
- Türk Telekom & ICT Authority 5G and Beyond Joint Graduate Support Programme 2023 – 2024
- TÜBİTAK Directorate of Science Fellowships and Grant Programme 2021 – 2023
- National ranks: **373rd** of 2M in University Entrance Exam; **11th** of 0.2M in ALES (National GRE)

SERVICE & TEACHING

- **Conference Reviewer:** ICLR '25–'27, NeurIPS '26, AISTATS '26–'27, AAAI '25
- **Journal Reviewer:** IEEE Trans. Artificial Intelligence '25, IEEE Internet of Things Journal '24–'25
- **Teaching Assistant, Bilkent University** (2021 – 2024): Signals and Systems, Probability and Statistics